NAME - ANSHIKA KASHYAP
— 2401010021
A
B.tech CSE (CORE)

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DBMS ASSIGNMENT.

Ques-1 What is SQL? why is it important in DBMS?

Ans-1 SQL (Structured Query Language) is a standard language used to create, manage and retrieve data.

- Used to perform operations like CREATE, INSERT, UPDATE, DELETE
- Helps in data retrieval using queries.
- Ensures efficient data management.
- Works with like MySQL, SQL Server

Ques-2 What is the different between DDL and DML?

Ans-2

DDL	DML
Data Definition language	Data Manipulation language
Defines structure	Manipulates data
CREATE, ALTER, DROP	INSERT, UPDATE, DELETE
Changes table structure	Changes data inside table

DDL → Structure
DML → Data

Ques-3 What is JOIN? Explain its types.

Ans-3 A JOIN is used to combine data from two or more tables based on a related column.

Types of JOIN -

INNER JOIN - returns matching records
LEFT JOIN - all records from left table + matching from right.
RIGHT JOIN - all records from right table + matching from left
FULL JOIN - all records from both tables.

Ques-4 What is GROUP BY and HAVING Clause?
Ans-4 GROUP BY → Groups rows having same values
HAVING → filters grouped data

Ques-5 What is Subquery?
Ans-5 A subquery is a query inside another query used to perform complex operations

- Also called nested query.
- Used in WHERE, FROM, SELECT
- Helps in filtering based on another query result

Ques-6 What is Relational Algebra?
Ans-6 Relational Algebra is a procedural query language used in DBMS

- Selection (σ) → filter rows
- Projection (π) → select columns
- Join ($\bowtie$) → combine tables
- Union, Intersection, Difference

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Ques-7 Difference b/w where and having?

WHERE	HAVING
Filter rows	filter groups
Before GROUP BY	Applied after GROUP BY
works on Individual rows	works on aggregated data
Aggregate function not used	used (SUM, AVG) etc.

SECTION-B (Query)

Ques-1 Write an SQL query to retrieve all students with marks > 75.

```
SELECT * FROM STUDENT
WHERE marks > 75;
```

SELECT → to fetch data
WHERE → to filter condition

Ques-2 Write a query using JOIN to combine Student and Courses tables.

```
SELECT s.student name, c.course_name FROM STUDENT s
JOIN COURSE c ON s.course_id = c.course_id;
```

JOIN = combines two tables
ON = condition to match records.

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Que-3 Write a query using GROUP BY to count students per department

SELECT department, COUNT(*) AS total_students FROM STUDENT
GROUP BY department;

GROUP BY : groups by department
COUNT() : counts students

SECTION-C : (Case-based)

Que-1 A company database stores employee and department data. Write a query to find employees in a specific department.

Ans-1
```
Create database company;
Use company;
CREATE TABLE EMPLOYEES
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50);
    emp_salary decimal(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFRENCES DEPARTMENS (dept_no)
);
CREATE TABLE DEPARTMENTS
    dept_no INT,
    dept_name INT PRIMARY KEY,
);
```

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1. Find employees in a specific department

```
SELECT e.emp_name, d.dept_name FROM EMPLOYEES e
JOIN DEPARTMENTS d ON e.dept_no = d.dept_no
WHERE d.dept_name = 'sales';
```

JOIN - combine employees & department
WHERE - filter specific department

2. Retrieve total sales per customer

```
SELECT customer_id, SUM(amount) AS total_sales FROM ORDERS
GROUP BY customer_id;
```

GROUP BY - group by customer
SUM() - total sales.

3. Find top 3 students

```
SELECT student_name, marks FROM STUDENT
ORDER BY marks DESC LIMIT 3;
```

ORDER BY - highest marks first
LIMIT - top 3 result

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SECTION - D (One Query + One Insight)

Ques-1 Write an SQL query for any real-life system and explain logic.

Ans-1 Scenario: Banking System
Find customers with balance > 50,000

```
SELECT customer_name, balance FROM BANK_ACCOUNTS
WHERE balance > 50,000;
```

Logic:
- SELECT retrieves required columns
- WHERE filters customers with high balance
- Used in real-life to identify premimum customers for special services.

Quest-2 Write a JOIN query and mention one use case.

Ans-2 Scenario: Employee - Department Relationship.

```
SELECT e.emp_name, d.dept_name FROM EMPLOYEES e
JOIN DEPARTMENTS d ON e.dept_no = d.dept_no;
```

Logic:
- JOIN combines data from multiple tables
- Used in real systems like:
  - ↳ HR systems → show employee with department
  - ↳ Company dashboards → organizational structure

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Write a GROUP BY query and explain its output.

Scenario : Count students per department

```
SELECT department, COUNT(*) AS total_students FROM STUDENT,
GROUP BY department;
```

Insight

- GROUP_BY groups students by department
- COUNT( ) counts number of students in each group.
- Output shows total students per department